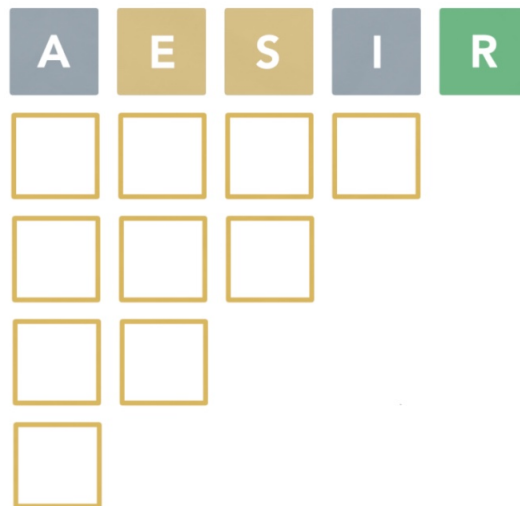


Hacking Wordle: A CS Perspective on the Viral Word Game

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1 Introduction: What Is It?

Does Josh Wardle's Wordle need any introduction? The simple word game has spread like wildfire, attracting millions of players across the world at its peak in early 2022. Every day, the website chooses a random five letter word that the player tries to solve for in as few guesses as possible. After every guess of a letter, the color of the five background tiles changes to grey, yellow, or green, indicating how close they were to the chosen word. It is grey if the letter is not in the solution word, yellow if the letter is in the solution word but in the wrong position, and green if the letter is present and in the right place. The player can keep guessing until they have either guessed the chosen word or used all six tries.

2 What is the Best Strategy?

With luck, any strategy can be considered the best strategy. If the starting word suggested by the strategy happens to be the solution word, then the strategy is technically proven to be effective.

3 The Worst Case

However, the lucky case, or optimal outcome for a strategy, is not helpful when evaluating how well one will perform when it is actually used, as it is almost always an edge case that occurs with

very low probability. More insightful measures include of how many guesses the average game of Wordle will take, or how many guesses the longest game of Wordle could take. For players never wanting to lose, the worst case, where they use up all their chances to guess the word, is most important.

Wordle has a list of roughly 10,000 valid guess words and a separate list of roughly 2,300 possible solution words, all of which are guessable. The guess words tend to be less familiar while the solution words are more common. When players guess a possible solution word, they get feedback that allows them to eliminate some of the possible solution words. Players win only when they guess the solution word. In the worst case, players would have to eliminate all but a single word with previous guesses to win.

A succinct way to represent this system is by associating each word with a “bucket” of other words. The bucket has all of the words which are still possible after that word is guessed. For example, if the possible solution words had been narrowed to “power,” “mower,” and “mover,” and the guessed word was “tower,” the associated bucket would be the list of two words “power” and “mower.” The game of Wordle can then be described as having a starting bucket of all the possible solution words, and then replacing that starting bucket with buckets associated with guesses until a bucket associated with a guess has only one word in it. That one word is the solution word.

4 An Upper Bound on Number of Guesses

Intuitively, a good strategy should eliminate many words with each guess, with the goal being able to choose guesses with small associated buckets. One such strategy is to greedily choose the guess which, in the worst case, would have the smallest bucket. For each possible remaining solution word, assume for a moment it is the current solution. Then find the bucket associated with each guess. This creates a list of possible buckets for each possible guess, each bucket assuming a different solution word. The heuristic for trying to minimize the worst case based on the above intuition is to eliminate as many words as possible with each guess, so the strategy should be to guess the word whose maximum bucket over all the possible solution words is minimum.

Simulating this strategy with a computer gives “aesir” as a starting word, which surprisingly matches the intuition of guessing words with common vowels first. Testing every possible solution word shows that the longest running game takes five guesses. This greedy strategy will make sure one never loses a game of Wordle!

This puts an upper bound on the amount of moves needed to guess a solution word in the worst case: five. But as the greedy solution is only a heuristic for a good strategy, it says nothing about a lower bound for the worst case number of needed guesses, which is definitely a larger computation.

5 A Lower Bound on Number of Guesses

A game of Wordle can further be represented by a path through a tree. The root of the tree is a bucket containing all the possible solution words. The edges represent guesses and connect nodes representing buckets before and after a certain guess. The size of the tree grows exponentially as longer paths of guesses are considered, quickly becoming too hard to manage.

Fortunately, considering long paths is irrelevant, since the point of searching this tree is to figure out if it is possible to solve every Wordle puzzle in four or fewer guesses. It is already known to be possible in five through the greedy strategy. Therefore, any path with more than four edges, a game where the player guesses more than four times, can be ignored.

The search space is still large, though now not so large that it cannot be searched with brute force. A recursive search, using the greedy strategy above as a heuristic to find solutions to words more quickly, will show that solving every Wordle puzzle with only four guesses is not possible; a lower bound for the worst case number of guesses is five.

This shows the greedy strategy is optimal for minimizing guesses in the worst case, but it is not the only one. Many strategies can achieve optimal performance, some being more possible for use by humans than others. Even a simple list of starter words can be effective, though at that point the problem becomes a test of human skill more than anything else.

6 A Quick Note on Greedy Adversarial Wordle

The strategies above are for the player to make the game as short as possible when Wordle happens to choose the word that forces the game to run as long as possible. But this brings about the question of what an adversarial game, a game trying to force the player to make as many guesses as possible, would look like.

An adversarial game which forces five guesses from the greedy strategy to be confident in a solution can always have the solution word “badly.” Always having the same word won’t work against every strategy though, since the strategy of always guessing “badly” quickly becomes very ineffective.

Instead, making a game which adapts to guesses is more versatile. The game will appear to have a set solution word, but in actuality it changes the solution word to prolong the game as the player guesses.

An example is a game which works like the greedy guessing strategy. It will have a bucket of possible solution words which it can choose from. When a player guesses, the bucket is split, and the part of the bucket containing the remaining possible solution words depends on the game’s response of greys, yellows, and greens. Analogous to how the greedy guessing strategy would choose the smallest bucket, the game would choose the response causing the current bucket to be as large as possible, helping to narrow to the least set of possible solutions.

While this creates a hard game to beat as a human, it is not optimal in prolonging guesses. The best it can do is force a four-word solution, for example, the sequence of guesses “aiery,” “canst,” “dolma,” or “offal.” As the game described above is deterministic, the sequence of words will always win.

The reason players cannot win in three guesses is because for them to be able to win in three guesses, they would have to narrow the word down to a single word by the time of their second guess. In response to each guess, the adversarial game divides the current bucket of words into at most $3^5 = 243$ smaller buckets; one bucket for each possible response. Each of these buckets

contains distinct words, so for there to only be one word in every bucket after the second guess, the maximum bucket size after the first guess would have to be at most 243 words. All guesses which sufficiently narrow the possible solutions do not lead to a three-word win. They are found through simply trying every possible pair of words after the first guess.

7 Conclusion

While the worst case is interesting, the most useful question for playing the game is: Which algorithm minimizes the average number of guesses? Strategies for optimizing the average case are endless, and writing up those strategies is similarly endless, so they won't be found here, though it is a worthwhile problem to think and read about. It might even be fun to take a crack at the daily Wordle today. Good luck!

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